SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I B. Sreesha 192126003 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Sreesha

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	5
Experimental Design	30	4
Use of Tools	20	5
Analysis of Results	15	4
Documentation	10	6

Total Marks 24 **/30**

Signature of the Faculty

SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Artificial Intelligence for Network Traffic Management

CO1 AT3 - TECHNICAL PRESENTATION

CSA0710 - COMPUTER NETWORKS

GUIDED BY

Dr. Shankar R,

Professor,

SIMATS Engineering

PRESENTED BY

SREESHA.B

1921260034

Introduction & Need for AI

- Network traffic management monitors, controls and optimizes data flow across a network.
- Modern traffic is huge and unpredictable: video streaming, cloud, IoT, 5G and remote work.
- Traditional static rules and manual configuration cannot react fast enough.
- AI learns from traffic data to predict, decide and adapt automatically in real time.

Why AI?

- Faster decisions
- Predicts congestion
- Detects threats early
- Scales with network size

How AI Manages Network Traffic

1

Collect Data

Packets, flows, latency, bandwidth, logs

2

Train Model

Learn normal and peak traffic patterns

3

Analyze

Classify traffic, predict load, spot anomalies

4

Act

Reroute, prioritize, allocate bandwidth

A continuous feedback loop: results of each action become new training data.

Key AI Techniques Used

Machine Learning

Decision trees, SVM, random forest classify traffic and flag anomalies.

Deep Learning

LSTM and CNN models forecast traffic and detect complex attack patterns.

Reinforcement Learning

Agents learn the best routing and congestion control by trial and reward.

Predictive Analytics

Time-series forecasting plans capacity before peak load occurs.

Applications

1. Traffic classification and identification
2. Congestion prediction and control
3. Intelligent load balancing and routing
4. Anomaly and intrusion detection
5. QoS and bandwidth allocation
6. Self-optimizing SDN, 5G and IoT networks

Benefits & Challenges

Benefits

- Real-time, automated decisions
- Better bandwidth utilization
- Lower latency and fewer outages
- Stronger network security
- Reduced operational cost

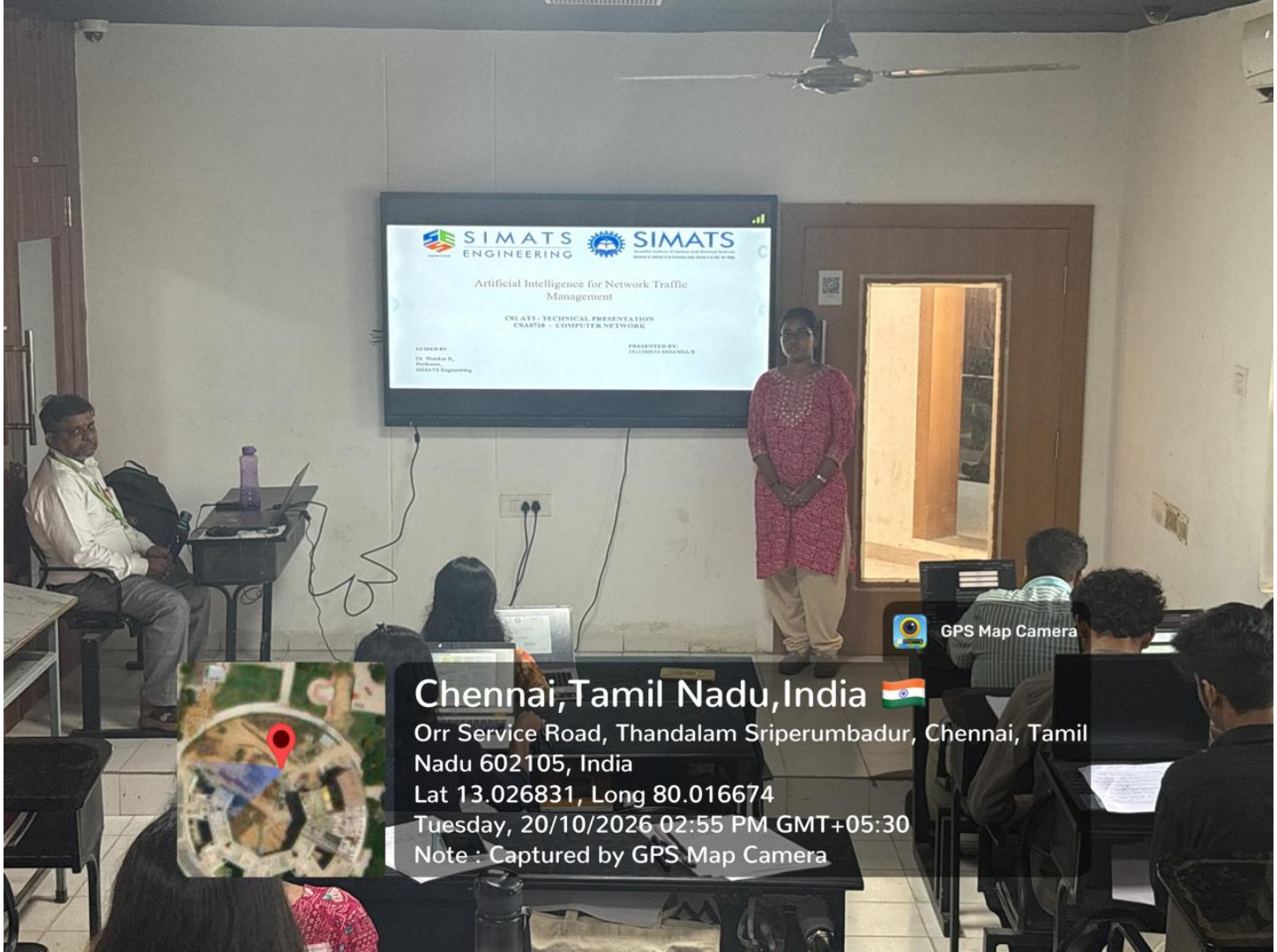
Challenges

- Needs large, quality training data
- High computation requirements
- Hard to explain model decisions
- Privacy and adversarial attack risks
- Integration with legacy networks

Conclusion & Future Scope

- AI turns network traffic management from reactive to proactive and self-optimizing.
- It improves performance, reliability, security and user experience.
- Future: self-healing networks, intent-based networking and AI-native 6G.
- Future: federated learning and explainable AI for trusted, privacy-safe operations.

Thank you


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Artificial Intelligence for Network Traffic Management
CS1 AT3 - TECHNICAL PRESENTATION
CSA0710 - COMPUTER NETWORK
GPS Map Camera
Chennai,Tamil Nadu,India
Orr Service Road, Thandalam Sriperumbadur, Chennai, Tamil Nadu 602105, India
Lat 13.026831, Long 80.016674
Tuesday, 20/10/2026 02:55 PM GMT+05:30
Note : Captured by GPS Map Camera

TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	Basic analysis with limited insights	Poor or incorrect analysis of results

			interpretation		
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation